CSP-203: Software Tools & Technologies Laboratory

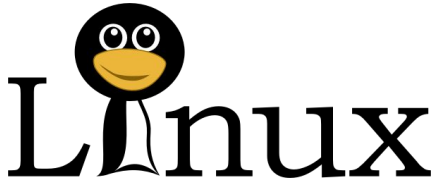


Lab Session - 3

Text Editing & Processing

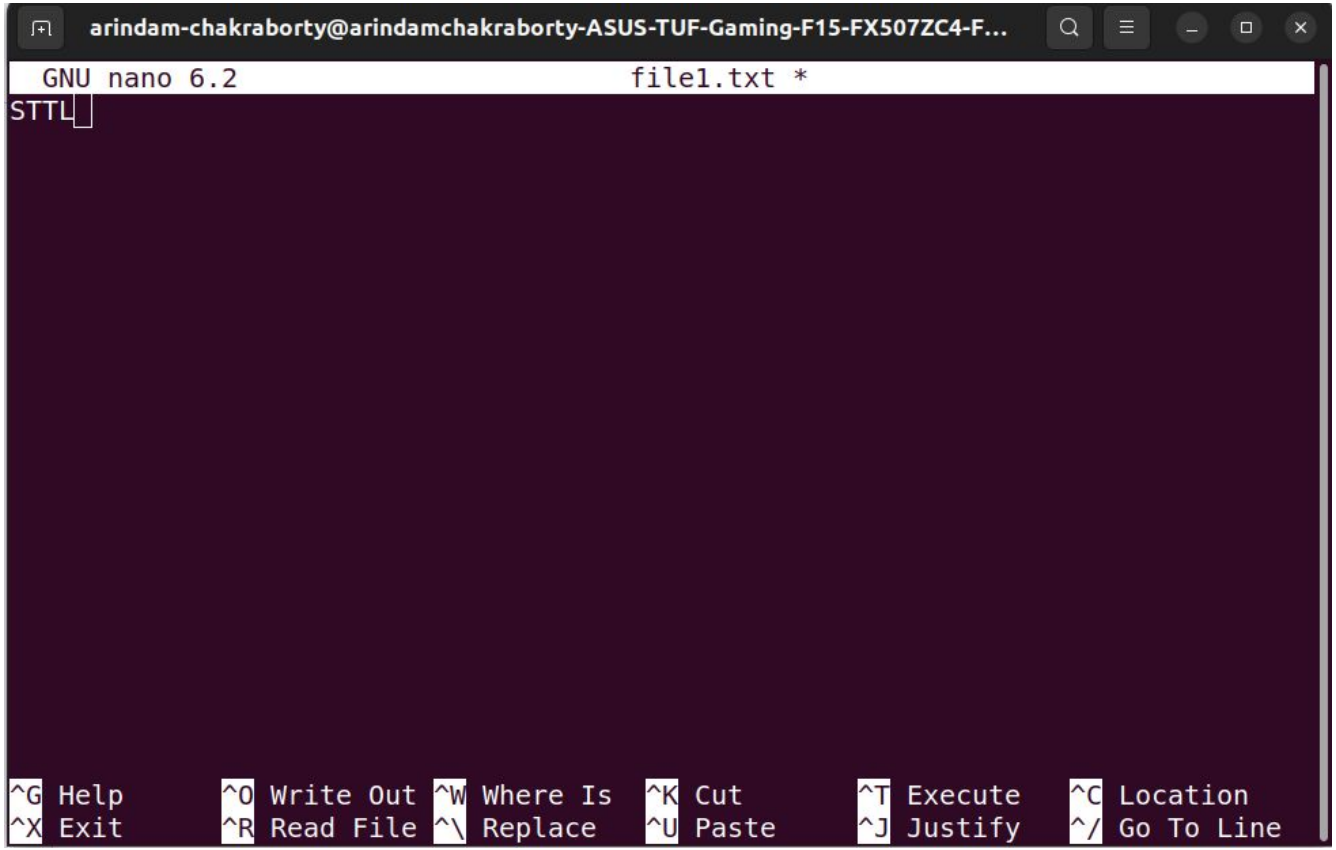
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Objectives:



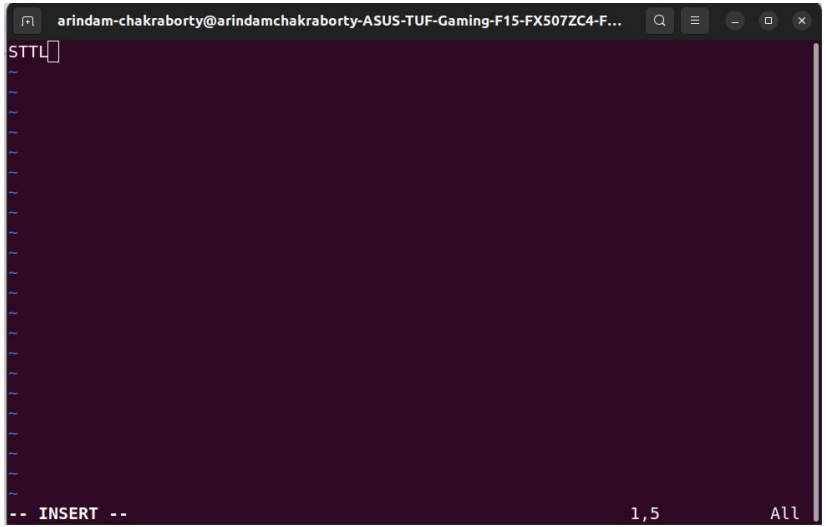
- Text Editors
- Text Processing Tools
  - Comparing Two Files
  - Viewing File
  - Counting Words, Lines, Bytes of File
  - Merging Files
  - Joining Files
  - Removing sections from file
  - Pattern Matching
  - Stream editor for filtering and transforming text
- Piping
  - Executing two or more commands at the same time

Text Editor (Nano):



More Nano Shortcuts: https://www.nano-editor.org/dist/latest/cheatsheet.html

Text Editor (Vim):



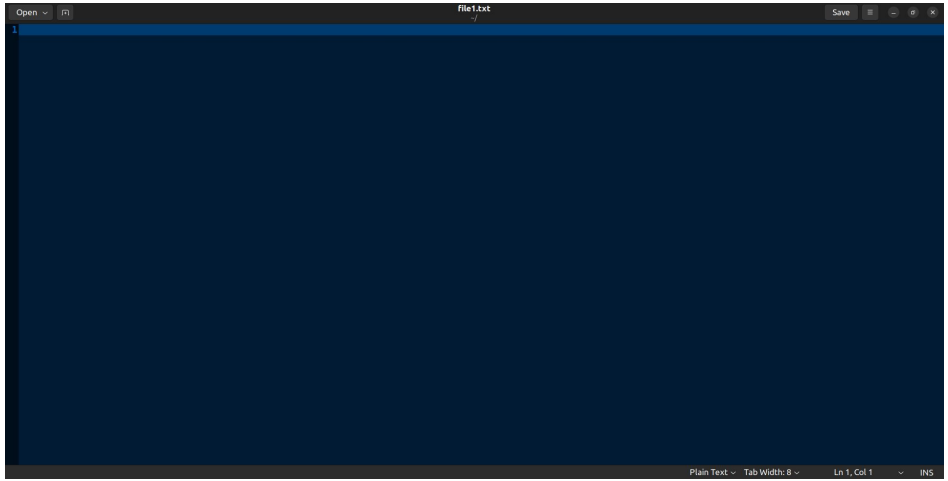
Three modes of vim editor:

1. Normal Mode (<ESC>)
2. Insert Mode (<i>)
3. Visual Mode (<v>)

Basic Vim Commands:

1. Save a file (:w)
2. Exit a file (:q)
3. Quit without exiting (:q!)

Text Editor (Gedit):



GNOME Editor

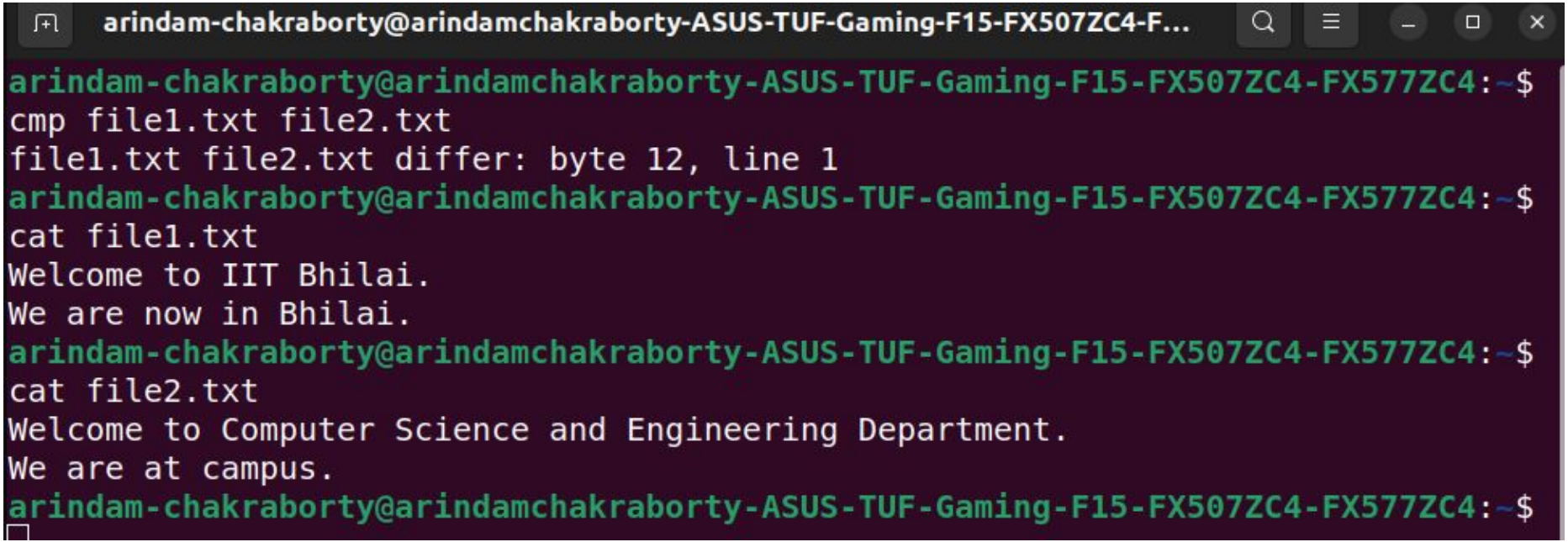
- Most user friendly, user interactive editor in linux.
- Modification to a file can be done easily.
- But Vim and Nano are useful while using ssh

Text Processing Tool (Comparing Two Files):

Compares two files byte by byte. (Command: cmp <file1> <file2>)

➔ Produces no output.

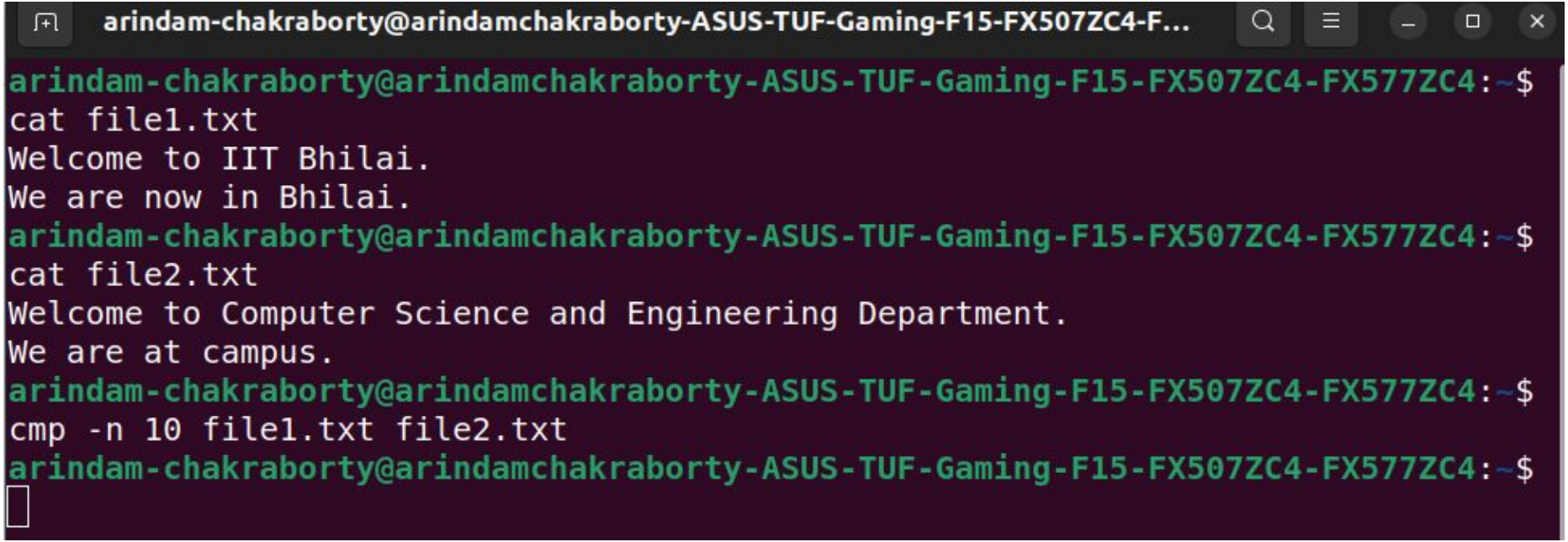
➔ Returns first byte having difference if they are not identical.



Text Processing Tool (Comparing Two Files):

Command: cmp -n <byte_no> <file1> <file2>

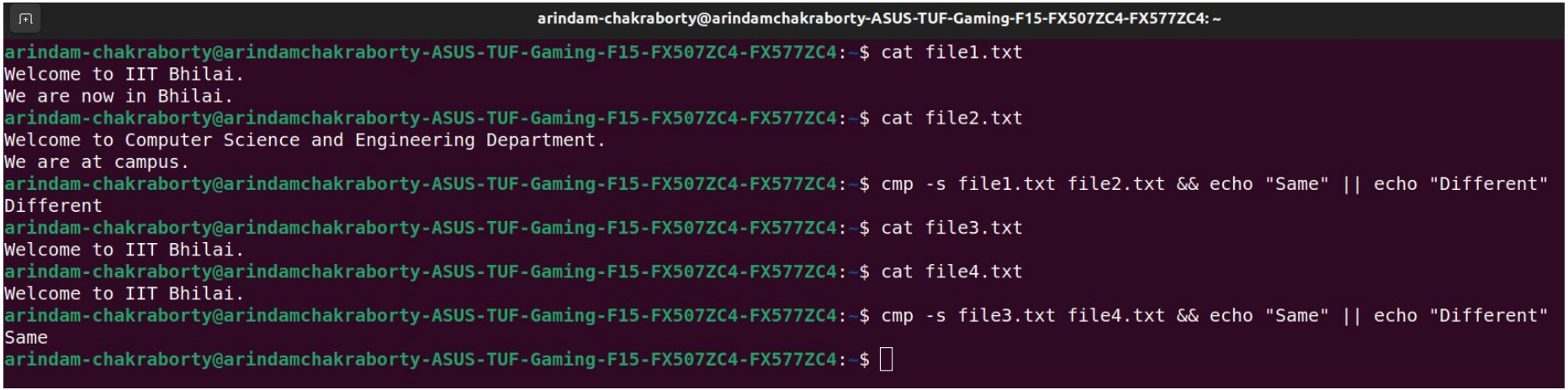
➔ Compares the file to a specific byte.



Text Processing Tool (Comparing Two Files):

Command: cmp -s <file1> <file2> && echo “Same” || echo “Different”

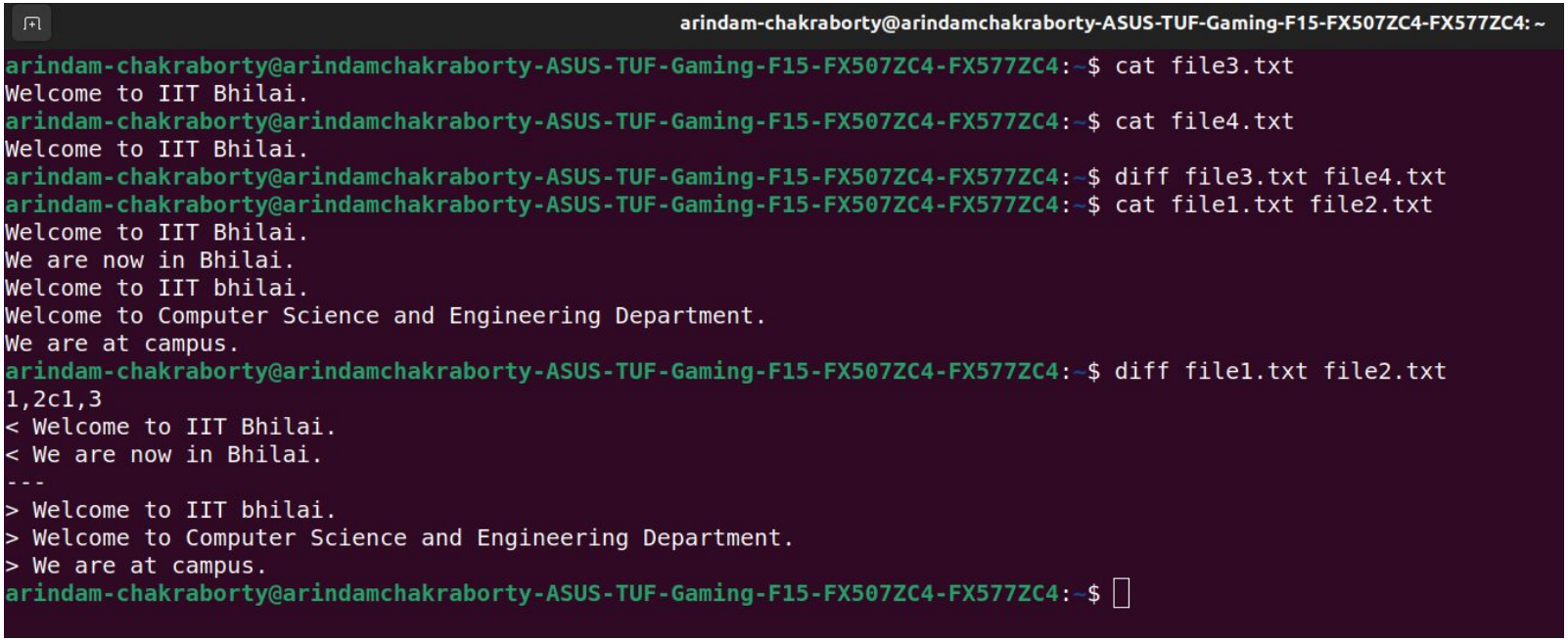
➔ -s silence the normal output of the command.



Text Processing Tool (Comparing Two Files):

Command: diff <file1> <file2>

➔ Compares two files line by line



Text Processing Tool (Comparing Two Files):

Command: diff -i <file1> <file2>

➔ Ignores case difference

```
> We are at campus.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ diff -i file1.txt file2.txt
2c2,3
< We are now in Bhilai.
---
> Welcome to Computer Science and Engineering Department.
> We are at campus.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$
```

➔ Command: diff -u <file1> <file2> [Unified format comparison]

➔ Command: diff -q <file1> <file2> [Only tells if they differ or not]

➔ Command: diff -y <file1> <file2> [Side by side comparison]

Text Processing Tool (Scroll up and Down):

Command: more <filename>

➔ Let scroll up and down through the file in the current terminal window.

Command: less <filename>

➔ Let view file the file in the form of pages.

Text Processing Tool (Word Count Command):

Command: wc <filename>

➔ Prints details of number of lines, number of words and size in bytes.

Command: wc -l <filename>

➔ Prints number of lines.

Command: wc -m <filename>

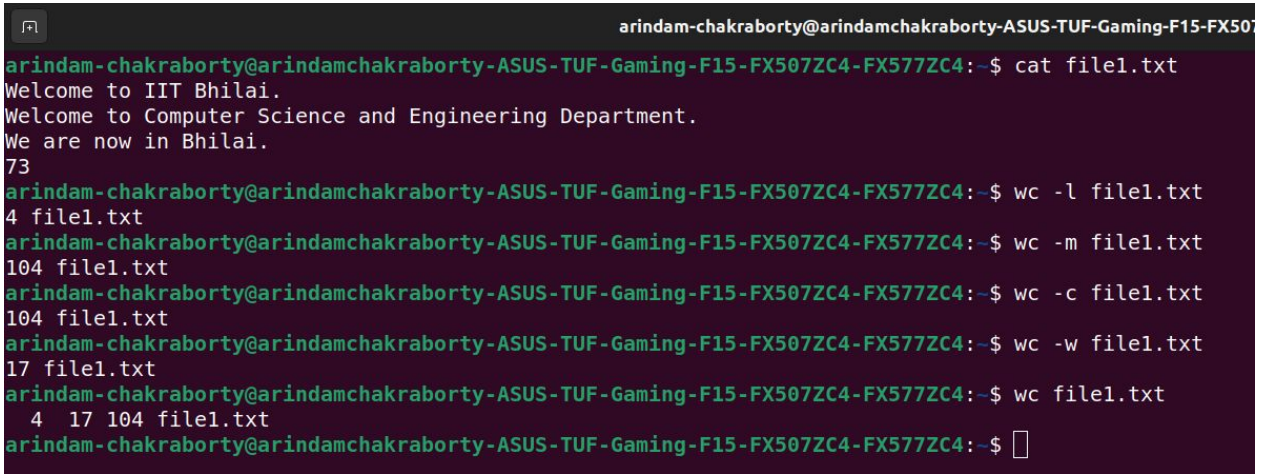
➔ Prints number of bytes.

Command: wc -c <filename>

➔ Prints number of characters.

Command: wc -w <filename>

➔ Prints details of number of words.



Text Processing Tool (Sorting command):

Command: sort <filename>

➔ Sort the lines of a file.

Command: sort -n <filename>

➔ Sort according to the string numeric value.

Command: sort -r <filename>

➔ Reverse the result of comparisons

Command: sort -g <filename>

➔ Sort according to the general numeric value.

Command: sort -f <filename>

➔ Ignore casing while comparison.

Command: sort <filename> -o <filename>

➔ Write the result to a file instead of std output.

➔ sort -t DELIM -k FIELD_NUM <filename> : sort based on specific field.

Text Processing Tool (Sorting command):

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ cat file6.txt
101, Student1
105, Student5
106, Student4
102, Student3
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ sort -n file6.txt
101, Student1
102, Student3
105, Student5
106, Student4
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ cat file8.txt
apple
Apple
banana
Banana
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ LC_ALL=C sort file8.txt
Apple
Banana
apple
banana
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$
```

Text Processing Tool (Cut Command):

Command: cut -b <specified_byte> <filename> / cut -c <specified_character> <filename>

➔ Shows only specified byte or specified characters from each line.

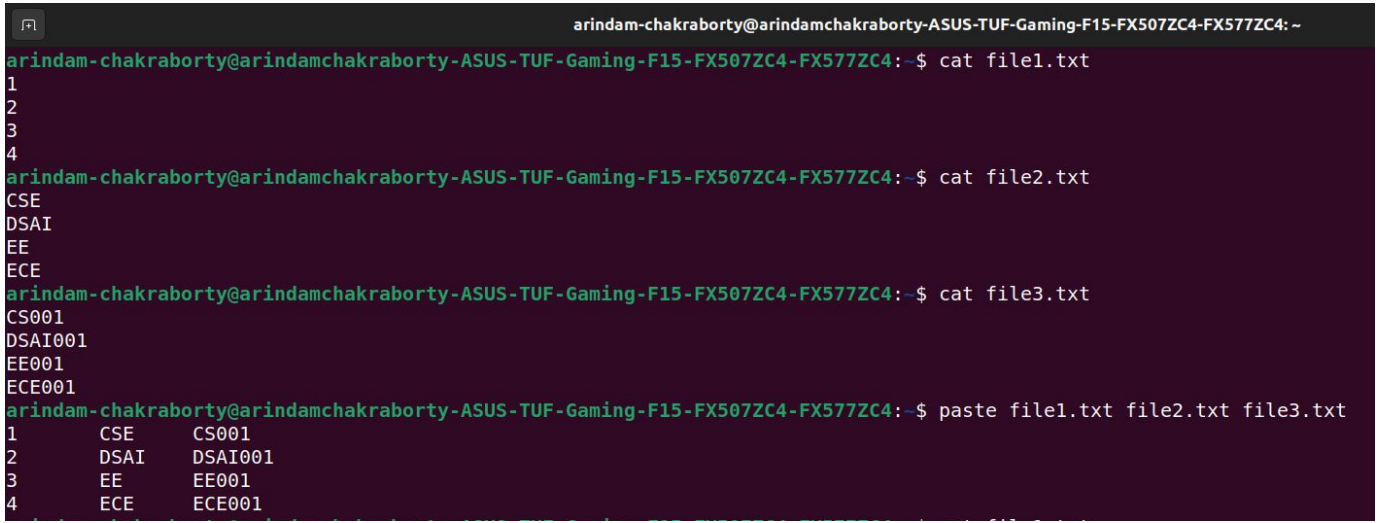


Text Processing Tool (Paste Command):

Command: paste <file1> <file2> … / paste -d ‘,’ <file1> <file2> …. [Custom delimiter]

➔ Joins multiple lines from files, uses TAB as default delimiter. Opposite of cut.

Command: paste -s <file1> <file2> ….. [Joining lines serially]

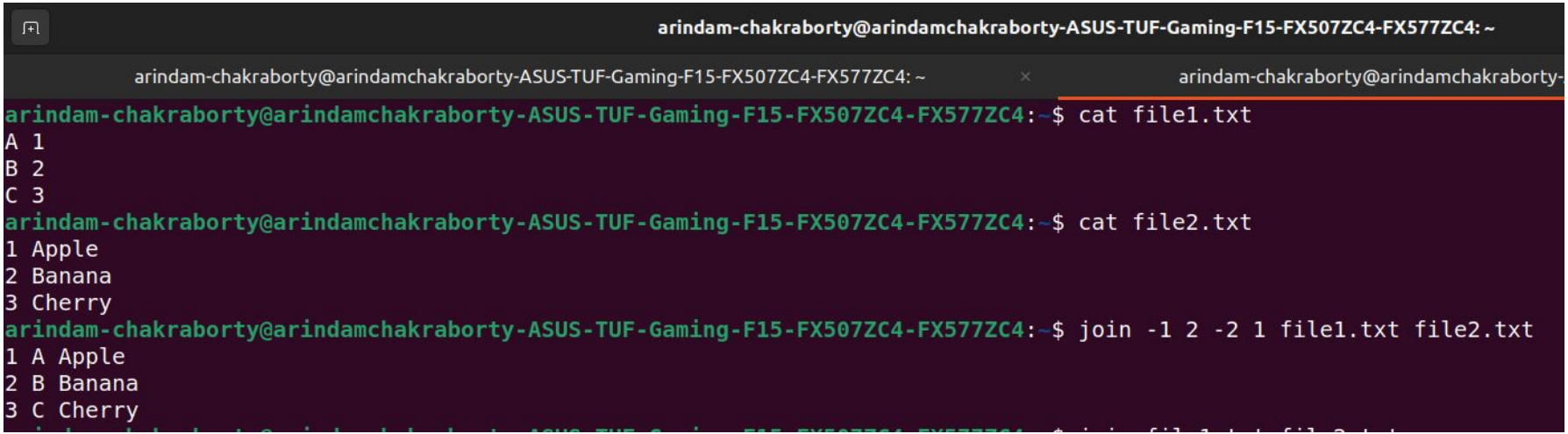


Text Processing Tool (Join Command):

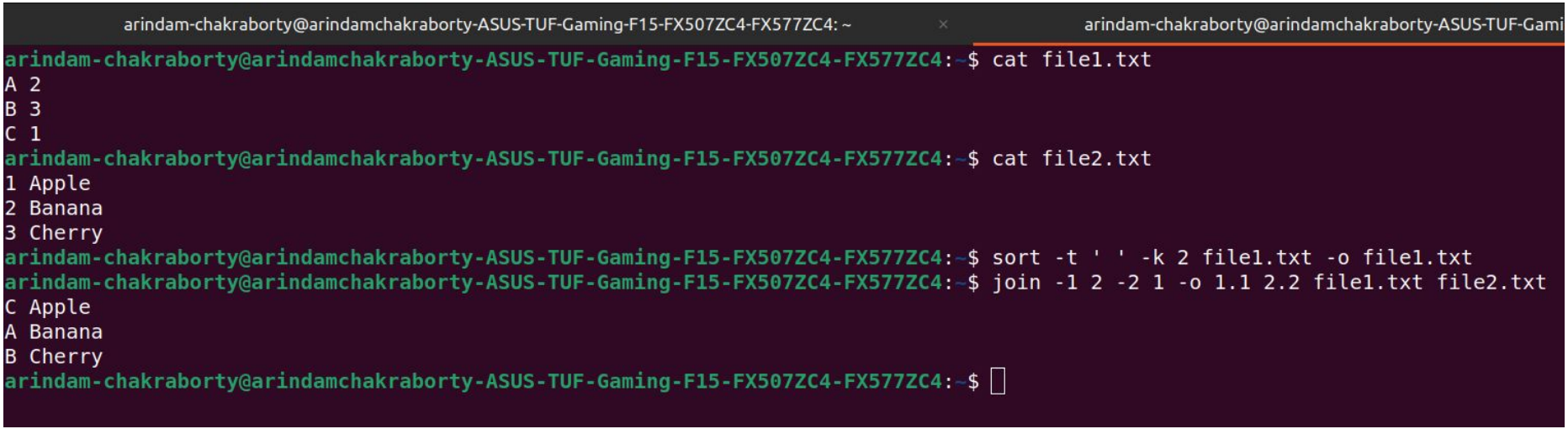
Command: join -1 <field_no> -2 <field_no> <file1> <file2>

➔ Joins two files based on specific field.

➔ join -1 <field_no> -2 <field_no> -o 1.<fieldno>... 2.<fieldno>... <file1> <file2>



Specifying delimiter: join -t “DELIM” ……..


arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4: ~
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gami
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ cat file1.txt
A 2
B 3
C 1
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ cat file2.txt
1 Apple
2 Banana
3 Cherry
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ sort -t ' ' -k 2 file1.txt -o file1.txt
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ join -1 2 -2 1 -o 1.1 2.2 file1.txt file2.txt
C Apple
A Banana
B Cherry
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$

Text Processing Tool (Grep Command):

- ➔ Command: grep “string” <file_name> [Highlights all the substrings matching the given string and outputs the line]
- ➔ Command: grep -i “string” <file_name> [Ignores case matching]
- ➔ Command: grep -n “string” <file_name> [Gives the line and corresponding line number]
- ➔ Command: grep -r “string” <directory_name> [Recursively search a directory for a given string]
- ➔ Command: grep -w “string” <file_name> [Gives lines with exactly matching words, not substring]
- ➔ Command: grep -v “string” <file_name> [Give non-matching lines]
- ➔ Command: grep -m <num> “string” <file_name> [Print only first num lines with matching substring]
- ➔ Command: grep -c “string” <file_name> [Give number of lines matching substring]
- ➔ Command: grep -o “string” <file_name> [Shows only part of the matching]
- ➔ Command: grep -E “pattern” <file_name> [Basic regular expression]
- ➔ Command: grep –x “string” <file_name> [Match only whole line]

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ cat file10.txt
Welcome to IIT Bhilai.
This is Computer Science and Engineering Department.
You are now at the campus.
Campus is very large.
You will be staying at the campus.
Campus10 version of the file has been createed.
Campus11 version has not been created yet.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ grep "campus" file10.txt
You are now at the campus.
You will be staying at the campus.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ grep "Campus" file10.txt
Campus is very large.
Campus10 version of the file has been createed.
Campus11 version has not been created yet.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ grep -i "campus" file10.txt
You are now at the campus.
Campus is very large.
You will be staying at the campus.
Campus10 version of the file has been createed.
Campus11 version has not been created yet.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ grep -n "campus" file10.txt
3:You are now at the campus.
5:You will be staying at the campus.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ cd ..
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:/home$ grep -r "campus" file10.txt
grep: file10.txt: No such file or directory
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:/home$ grep -r "campus" arindam-chakraborty
arindam-chakraborty/Downloads/mpquic-ns3-mpquic-1.2/src/mpi/examples/nms-p2p-nix-distributed.cc: * The fundamental unit of the NMS model consists of a campus
network. The
arindam-chakraborty/Downloads/mpquic-ns3-mpquic-1.2/src/mpi/examples/nms-p2p-nix-distributed.cc: * campus network topology can been seen here:
arindam-chakraborty/Downloads/mpquic-ns3-mpquic-1.2/src/mpi/examples/nms-p2p-nix-distributed.cc: * number of these campus networks can be connected together (
default 2)
arindam-chakraborty/Downloads/mpquic-ns3-mpquic-1.2/src/mpi/examples/nms-p2p-nix-distributed.cc:  cmd.AddValue ("campuses", "Number of campus networks", nCN);
arindam-chakraborty/Downloads/mpquic-ns3-mpquic-1.2/src/mpi/examples/nms-p2p-nix-distributed.cc:  cmd.AddValue ("single", "Use single on/off app per campus ne
twork", single);
arindam-chakraborty/Downloads/mpquic-ns3-mpquic-1.2/src/nix-vector-routing/examples/nms-p2p-nix.cc: * The fundamental unit of the NMS model consists of a camp
```

❑ Regular Expression Basic:

Operator	Meaning	Example	Matches
.	Any single character	a.b	aab, acb
^	Start of string	^abc	abc, but not xabc
$	End of string	abc$	abc, but not abcc
*	Zero or more of the previous character	ab*c	ac, abc, abbc
+	One or more of the previous character	ab+c	abc, abbc, but not ac
?	Zero or one of the previous character	ab?c	ac, abc
[]	Character class (one of)	[abc]	a, b, or c
[^]	Negated character class	[^abc]	Any character except a, b, or c

Text Processing Tool (Awk Command):

- awk 'pattern { action }' filename
  - pattern: A condition or regex to match lines.
  - action: Commands to execute when pattern matches.
  - filename: The file to process (can also use input redirection or piping).
    - $0: Entire line
    - $1, $2, ..., $n: Fields (words/columns) in the line
    - NF: Number of fields in current line
    - NR: Number of current record (line number)
    - FS: Field separator (default: space or tab)
    - OFS: Output field separator
    - RS: Record separator

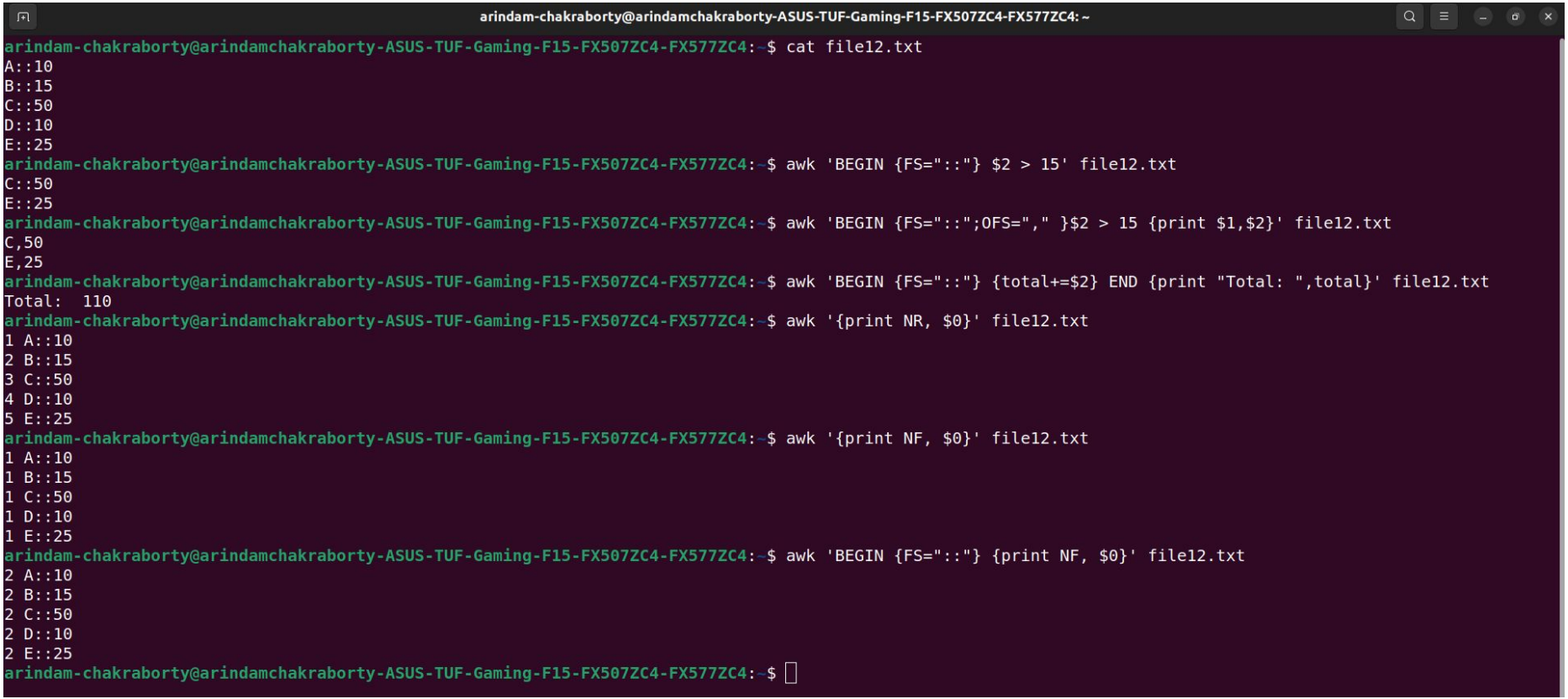
Text Processing Tool (Awk Command):

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ cat file11.txt
A 10
B 15
C 50
D 10
E 25
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk '{print $1}' file11.txt
A
B
C
D
E
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk '{print $1,$2}' file11.txt
A 10
B 15
C 50
D 10
E 25
```

arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4: ~

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ cat file12.txt
A 10
B 15
C 50
D 10
E 25
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk 'BEGIN {FS = " "}{OFS="::"} {print $1,$2}' file12.txt  > temp.txt && mv te
mp.txt file12.txt
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ cat file12.txt
A::10
B::15
C::50
D::10
E::25
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$
```

Text Processing Tool (Awk Command):



Text Processing Tool (Awk Command):

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk 'BEGIN {RS="#";FS=",";OFS="::"} {print "Record: ",$1,$2,$3}' file13.txt
Record: ::banana::mango::orange
Record: ::apple::guava::kiwi
```

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk -v num=5 'BEGIN {sum=num*(num+1)/2; print sum}'
15
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk -v num=5 'BEGIN {sum=0; for(i=1;i<=num;i++){sum+=i} print sum}'
15
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk 'BEGIN {sum=0; num=10; for(i=1;i<=num;i++){sum+=i} print sum}'
55
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk 'BEGIN {a=10;b=12; if(a>b){print "True"}else{print "False"}}'
False
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk 'BEGIN {a=18;b=12; if(a>b){print "True"}else{print "False"}}'
True
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk 'BEGIN {sum=0;num=10;while(num>0){sum+=num} print sum}'
^C
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk 'BEGIN {sum=0;num=10;while(num>0){sum+=num;num--} print sum}'
55
```

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ cat file12.txt
A::10
B::15
C::50
D::10
E::25
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk 'BEGIN {FS="::"}{if($2>15){print $1,$2,"Passed"}else{print $1,$2,"Failed"}
}' file12.txt
A 10 Failed
B 15 Failed
C 50 Passed
D 10 Failed
E 25 Passed
```

Text Processing Tool (Awk Command):

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ cat file12.txt
A::10
B::15
C::50
D::10
E::25
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk 'BEGIN {FS="::";OFS="->"} /^10$/ {print NR,$1,$2}' file12.txt
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk 'BEGIN {FS="::";OFS="->"} /^A::10$/ {print NR,$1,$2}' file12.txt
1->A->10
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk 'BEGIN {FS="::";OFS="->"} $2=="10" {print NR,$1,$2}' file12.txt
1->A->10
4->D->10
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$
```

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ cat file11.txt
A 10
B 15
C 50
D 10
E 25
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ awk '$2 ~ /^10$/ {print NR, $1,$2}' file11.txt
1 A 10
4 D 10
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$
```

Text Processing Tool (Sed Command):

- ➔ Sed stands for Stream Editor. It is a powerful Unix utility used to search, find, replace, insert, and delete text in a stream (like a file or input from a pipeline).
- ➔ Commands:
  - ◆ Replace first match per line: sed ‘s/<old_text>/<new_text>/ <filename>
  - ◆ Replace all matches per line: sed ‘s/<old_text>/<new_text>/g’ <filename>
  - ◆ Replace all matches per line (case insensitive): sed ‘s/<old_text>/<new_text>/Ig’ <filename>
  - ◆ Delete a line with a line number: sed ‘<line_num>d’ <filename>
  - ◆ Delete a line with given expression: sed ‘/expr/d’ <filename>
  - ◆ Make the changes permanent to the file: sed -i ‘.....’ <filename>
  - ◆ Create a backup of the file: sed -i.bak ‘.....’ <filename>

Text Processing Tool (Sed Command):

➔ Commands:

- Print a specific line number: sed -n ‘<line_number>p’ <filename>
- Print from line_n1 to line_n2: sed -n ‘<line_n1>,<line_n2>p’ <filename>
- Insert before line_num: sed ‘<line_num>i\<String>’ <filename>
- Append after line_num: sed ‘<line_num>a\<String>’ <filename>
- Replace line_num: sed ‘<line_num>c\<String>’ <filename>

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ cat file12.txt
This is line 1 and it is being edited.
This is line 2 and it is being edited.
This is line 3 and it is being edited.
This is line 4 and it is being edited.
This is line 5 and it is being edited.
That is line 6 and it is being edited.
That is line 7 and it is being edited.
That is line 8 and it is being edited.
That is line 9 and it is being edited.
That is line 10 and it is being edited.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ sed 's/is/are/' file12.txt
Thare is line 1 and it is being edited.
Thare is line 2 and it is being edited.
Thare is line 3 and it is being edited.
Thare is line 4 and it is being edited.
Thare is line 5 and it is being edited.
That are line 6 and it is being edited.
That are line 7 and it is being edited.
That are line 8 and it is being edited.
That are line 9 and it is being edited.
That are line 10 and it is being edited.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ sed 's/ is/ are/' file12.txt
This are line 1 and it is being edited.
This are line 2 and it is being edited.
This are line 3 and it is being edited.
This are line 4 and it is being edited.
This are line 5 and it is being edited.
That are line 6 and it is being edited.
That are line 7 and it is being edited.
That are line 8 and it is being edited.
That are line 9 and it is being edited.
That are line 10 and it is being edited.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ sed 's/ is/ are/g' file12.txt
This are line 1 and it are being edited.
This are line 2 and it are being edited.
This are line 3 and it are being edited.
This are line 4 and it are being edited.
This are line 5 and it are being edited.
```

```
That are line 10 and it are being edited.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ sed 's/that/those/Ig' file12.txt
This is line 1 and it is being edited.
This is line 2 and it is being edited.
This is line 3 and it is being edited.
This is line 4 and it is being edited.
This is line 5 and it is being edited.
those is line 6 and it is being edited.
those is line 7 and it is being edited.
those is line 8 and it is being edited.
those is line 9 and it is being edited.
those is line 10 and it is being edited.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$
```


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That are line 10 and it are being edited.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ sed 's/that/those/Ig' file12.txt
This is line 1 and it is being edited.
This is line 2 and it is being edited.
This is line 3 and it is being edited.
This is line 4 and it is being edited.
This is line 5 and it is being edited.
those is line 6 and it is being edited.
those is line 7 and it is being edited.
those is line 8 and it is being edited.
those is line 9 and it is being edited.
those is line 10 and it is being edited.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$
```

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arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4: ~
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ sed -n '2p' file12.txt
This is line 2 and it is being edited.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ sed -n '2,4p' file12.txt
This is line 2 and it is being edited.
This is line 3 and it is being edited.
This is line 4 and it is being edited.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ sed '2i\Before line 2' file12.txt
This is line 1 and it is being edited.
Before line 2
This is line 2 and it is being edited.
This is line 3 and it is being edited.
This is line 4 and it is being edited.
This is line 5 and it is being edited.
That is line 6 and it is being edited.
That is line 7 and it is being edited.
That is line 8 and it is being edited.
That is line 9 and it is being edited.
That is line 10 and it is being edited.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ sed '2a\After line 2' file12.txt
This is line 1 and it is being edited.
This is line 2 and it is being edited.
After line 2
This is line 3 and it is being edited.
This is line 4 and it is being edited.
This is line 5 and it is being edited.
That is line 6 and it is being edited.
That is line 7 and it is being edited.
That is line 8 and it is being edited.
That is line 9 and it is being edited.
That is line 10 and it is being edited.
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ sed '2c\Modified line 2' file12.txt
This is line 1 and it is being edited.
Modified line 2
This is line 3 and it is being edited.
This is line 4 and it is being edited.
This is line 5 and it is being edited.
That is line 6 and it is being edited.
That is line 7 and it is being edited.
That is line 8 and it is being edited.
```


THANK
YOU